

HEALTH CARE

Health care courses offer students an opportunity to investigate factors contributing to personal health and, at the same time, gain an introduction to a range of careers in the health care industry. These careers include, but are not limited to, child care worker, dental assistant/dentist, gerontologist, laboratory technician/technologist, doctor, nursing assistant/nurse, pharmacy assistant/pharmacist, and personal support worker. Courses in health care help prepare students for rewarding careers in a sector that is set to expand rapidly as our population ages.

Students will gain hands-on experience using industry-standard instruments, equipment, and materials and practising current techniques. They will learn the professional terminology of the field, acquire transferable problem-solving skills, and expand their communication and interpersonal skills as they interact with their peers and clients in a variety of simulated care scenarios. As they acquire theoretical understanding and practical skills in their areas of interest, students will become better prepared to make informed career choices.

The list of approved emphasis areas for health care can be found at www.edu.gov.on.ca/eng/curriculum/secondary/teched.html.

- Courses in technological education are suitable for use in cooperative education programs and in connection with other forms of experiential learning as well as in programs such as the Specialist High Skills Major (SHSM). For more information, see pages 43–44 of this document.
- For policy guidelines pertaining to multiple-credit courses and emphasis courses, see pages 17–18 of this document.

Health Care: Support Services, Grade 12

Workplace Preparation

TPJ4E

This course enables students to develop the basic skills needed for careers in a range of health care support services. Students will practise and apply a variety of clinical procedures and infection control skills as they learn about principles of infection control, service excellence, and the nature of the health care industry. Students will also investigate workers' health and safety issues, environmental and societal issues related to health care, and career opportunities in the field.

Prerequisite: None

A. HEALTH CARE FUNDAMENTALS

OVERALL EXPECTATIONS

By the end of this course, students will:

- A1.** demonstrate an understanding of health care terminology and its correct usage;
- A2.** demonstrate an understanding of the basic anatomy and physiology of the human body;
- A3.** identify relationships between lifestyle and the health of individuals;
- A4.** demonstrate an understanding of the chain of infection and practices for preventing the transmission of infection;
- A5.** compare conventional and complementary methods of disease prevention and treatment.

SPECIFIC EXPECTATIONS

A1. Health Care Terminology

By the end of this course, students will:

- A1.1** demonstrate an understanding of terminology related to health care concepts and procedures (*e.g., nothing by mouth/nil per os [NPO], rehabilitation, gerontology, acute, chronic*), and use it correctly in oral and written communication;
- A1.2** demonstrate an understanding of health care terminology describing parts of the human body and their location (*e.g., abdominal cavity, cranial cavity, posterior, superior*), and use it correctly in oral and written communication.

A2. Anatomy and Physiology

By the end of this course, students will:

- A2.1** identify the relationships between cells, tissues, organs, and systems in the structure of the human body (*e.g., different types of cells and cell arrangements in tissues, different types of tissues for different purposes in organs, combinations of organs working together in body systems*);
- A2.2** demonstrate a basic understanding of the anatomy and physiology of the human musculoskeletal system (*e.g., demonstrate types of movement for a variety of joints, such as flexion and extension of a hinge joint; identify major bones and muscles of the body*);

- A2.3** demonstrate a basic understanding of the anatomy and physiology of the human cardiovascular system (*e.g., by building a model of the heart to identify major structural features and blood flow*);

- A2.4** demonstrate a basic understanding of the anatomy and physiology of the human respiratory system (*e.g., by building a functional respiratory model to illustrate the mechanism of breathing*);

- A2.5** demonstrate a basic understanding of the anatomy and physiology of the human digestive system (*e.g., by developing a model illustrating the digestive pathway*).

A3. Lifestyle and Personal Health

By the end of this course, students will:

- A3.1** identify the impact of alterations in health (*e.g., hospitalization, long-term illness or disability*) throughout the lifespan (*e.g., impact on a toddler, teenager, elderly person*);
- A3.2** describe how lifestyle choices (*e.g., amount of sleep, dietary and exercise habits, substance use and abuse*) can affect an individual's health and well-being.

A4. Infection Control

By the end of this course, students will:

- A4.1** identify components of the chain of infection (*e.g., host, reservoir, mode of transmission, portal of entry*);
- A4.2** describe routine practices for preventing the transmission of infection (*e.g., isolation cleaning; droplet precautions; contact precautions; donning personal protective equipment such as gloves, masks, and gowns; disposal of body fluids*) and the indications for implementing them.

A5. Disease Prevention and Treatment

By the end of this course, students will:

- A5.1** compare conventional and complementary approaches to health care in terms of the therapeutic approaches used (*e.g., pharmaceutical medications versus herbal/natural remedies*) and the types of practitioners offering the services (*e.g., physicians versus homeopaths or Aboriginal healers*);
- A5.2** compare conventional and complementary treatment methods for a variety of health care issues (*e.g., nicotine patch versus acupuncture for cessation of smoking*).

B. HEALTH CARE SKILLS

OVERALL EXPECTATIONS

By the end of this course, students will:

- B1.** identify instruments, equipment, and materials that are commonly used in the health care industry, and use them correctly and safely;
- B2.** use vital signs to assess a person's health status;
- B3.** demonstrate the ability to apply health care skills and techniques safely and to industry standards;
- B4.** demonstrate the ability to apply a variety of techniques for communicating with clients and collecting client information.

SPECIFIC EXPECTATIONS

B1. Instruments, Equipment, and Materials

By the end of this course, students will:

- B1.1** identify and explain the purpose of common instruments, equipment, and materials used for client care in the health care industry (*e.g., patient lift, commode, intravenous [IV] pole, hospital bed, wheelchair*), and apply correct procedures for their use;
- B1.2** identify health care problems related to the use of instruments, equipment, or materials (*e.g., spread of infection through a health care facility as a result of the use of contaminated equipment*), and apply proper control procedures (*e.g., hand hygiene protocols, use of personal protective equipment guidelines, proper handling of bed linens*) to prevent their occurrence.

B2. Vital Signs

By the end of this course, students will:

- B2.1** demonstrate the ability to use instruments (*e.g., thermometer, sphygmomanometer, stethoscope*) to measure the four primary vital signs (temperature, pulse, respiration, and blood pressure);
- B2.2** assess and document a client's vital signs: temperature (by oral, axillary, and tympanic routes), pulse rate (by apical and radial sites), respiration rate, and blood pressure;

- B2.3** identify normal values for temperature, pulse, respiration, and blood pressure across the lifespan, and identify possible health implications of abnormal values (*e.g., elevated temperature suggests a possible infection*).

B3. Skills and Techniques

By the end of this course, students will:

- B3.1** demonstrate an understanding of correct hand hygiene procedures (*e.g., knowing when they are required, knowing when to use hand washing rather than antibacterial sanitizers*), and apply as required;
- B3.2** make an unoccupied and occupied bed correctly, using correct principles of infection control;
- B3.3** perform mobility techniques (*e.g., turning, lifting, and transferring clients; using crutches or walkers and instructing clients in their use*) correctly and safely;
- B3.4** demonstrate an understanding of and apply proper body mechanics and ergonomics when performing health care procedures (*e.g., bend with the knees instead of the back, use a wide base of support when lifting or turning*);
- B3.5** demonstrate an understanding of and apply safe practices for handling, preparing, and storing food in a real or simulated care environment (*e.g., use proper cooking temperatures to kill bacteria; check that clients' food tolerances, requirements, or restrictions are observed*);

B3.6 demonstrate an understanding of and apply infection control skills in a simulated care environment (*e.g., isolation cleaning; droplet precautions; contact precautions; donning personal protective equipment such as gloves, masks, and gowns; disposal of body fluids*).

B4. Communication and Documentation

By the end of this course, students will:

- B4.1** demonstrate an understanding (*e.g., through role play*) of therapeutic communication techniques, including conflict-resolution skills;
- B4.2** identify various communication barriers (*e.g., barriers relating to health condition, age and stage of development, and differing cultural communication styles*), and apply strategies to overcome these difficulties.

C. HEALTH CARE, THE ENVIRONMENT, AND SOCIETY

OVERALL EXPECTATIONS

By the end of this course, students will:

- C1.** identify the impact of medical wastes on the environment, and describe ways of protecting the environment from these hazards;
- C2.** demonstrate an understanding of ways in which health care issues and societal issues are interrelated.

SPECIFIC EXPECTATIONS

C1. Health Care and the Environment

By the end of this course, students will:

- C1.1** identify the environmental impact of technological advances in the health care field (*e.g., single-use devices create end disposal issues, as incineration results in heavy metal toxicity and landfilling creates biohazardous leachate; because of the increased use of pharmaceuticals, traces of prescription drugs are now being found in surface water*);
- C1.2** research and report on laws, regulations, guidelines, and information sources pertaining to the disposal of medical waste (*e.g., Atomic Energy Control Board – radioactive materials; Workplace Hazardous Materials Information System [WHMIS] – chemical hazards; Ontario Regulation 102/94 – waste management*).

C2. Health Care and Society

By the end of this course, students will:

- C2.1** identify current societal trends or issues and their implications for health care (*e.g., lack of exercise leading to increased obesity and related health problems; aging of the population and increased need for geriatric care; global travel and increased prevalence of exotic diseases; cultural diversity and the increased complexity of delivering health care*);
- C2.2** identify current health care issues from recent media coverage and describe their societal implications (*e.g., outbreaks of potentially epidemic diseases and the need for social controls such as quarantine; challenges in providing basic health care resources in remote communities; funding limitations and costs of covering expensive treatments and drugs; antibiotic-resistant diseases, nosocomial infection*).

D. PROFESSIONAL PRACTICE AND CAREER OPPORTUNITIES

OVERALL EXPECTATIONS

By the end of this course, students will:

- D1.** demonstrate an understanding of laws and regulations governing the health and safety of workers in the health care industry;
- D2.** demonstrate an understanding of legal and ethical standards governing the practice of health care in Ontario;
- D3.** demonstrate an understanding of the principles of good customer service and client-centred care, and apply these principles in a simulated care setting;
- D4.** describe career opportunities in the health care field and their related educational requirements.

SPECIFIC EXPECTATIONS

D1. Health and Safety

By the end of this course, students will:

- D1.1** identify and describe legislation and sources of information pertinent to worker safety in the health care workplace (*e.g., Occupational Health and Safety Act [OHSA], Workplace Hazardous Materials Information System [WHMIS]*);
- D1.2** demonstrate an understanding of and apply safe procedures for the use of tools, materials, and equipment as outlined in safety manuals, operating instructions, and institutional requirements;
- D1.3** describe and apply safe methods for the handling, storage, and disposal of waste and biohazardous materials (*e.g., use of a sharps container*);
- D1.4** use protective clothing and equipment as required to keep themselves and others safe and free from harm.

D2. Professional Standards

By the end of this course, students will:

- D2.1** identify the principal components of legislation regulating the practice of health care in Ontario (*e.g., Regulated Health Professions Act [RHPA] sets the framework for regulating the scope of practice of health professions*);

- D2.2** describe the professional and ethical standards that health care workers are expected to adhere to (*e.g., accountability, knowledge, competence, respect for patients' rights and privacy, confidentiality, respect for cultural and religious diversity, respect for the sanctity of life*);

- D2.3** explain the importance of informed consent and informed decision making in client care;

- D2.4** describe what health care workers can do to project a professional image (*e.g., behave courteously at all times; respond to client inquiries promptly and accurately; be supportive; wear appropriate clothing and maintain a tidy personal appearance; wear a name badge and keep it visible at all times*), and explain why it is important to do so.

D3. Client-centred Care

By the end of this course, students will:

- D3.1** demonstrate an understanding of how to manage time and provide care efficiently in order to enhance client outcomes;
- D3.2** explain the importance of understanding client expectations and helping clients understand what they can reasonably expect of high-quality client service, and describe some of the benefits of working to exceed those expectations;
- D3.3** demonstrate an understanding of and apply effective service recovery techniques (*e.g., actively listen to complaints, offer blameless apologies, propose solutions*) to resolve client complaints and grievances.

D4. Career Opportunities

By the end of this course, students will:

- D4.1** identify career opportunities in the health care field and describe the education and/or training required for entry into these occupations;
- D4.2** describe the various types of community health and social services (e.g., *community care access centres, nursing, mental health services, physiotherapy, laboratory services, Meals on Wheels, vision care services*) available in their community and the career opportunities they provide;
- D4.3** demonstrate an understanding of and apply the Essential Skills that are important for success in the health care industry, as identified in the Ontario Skills Passport (e.g., *computer use, data analysis, job task planning and organizing*);
- D4.4** demonstrate an understanding of and apply the work habits that are important for success in the health care industry, as identified in the Ontario Skills Passport (e.g., *teamwork, organization, self-advocacy*);
- D4.5** maintain an up-to-date portfolio that includes pieces of work and other materials that provide evidence of their skills and achievements in health care (e.g., *Passport to Safety certificate, reports, assignments, Ontario Skills Passport Work Plan and Transition Plan*), and explain why having a current portfolio is important for career development and advancement.